



Atomic packing efficiency and phase transition in a high entropy alloy

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ABSTRACT

With the increasing of Al content in a high entropy alloy of $\text{Ti}_{0.5}\text{CrFeCoNiAl}_x\text{Cu}_{1-x}$, the phase structure of face centered cubic (FCC) transits to body centered cubic (BCC), namely from a structure with high atomic packing efficiency (APE) to a lower one, and the compressive strength increases from 1650 MPa to 2697 MPa. The phase transition and the strength increases are explained by the lattice distortion.

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1. Introduction

The alloys with more than five principle components and near equi-atomic ratio can be defined as high entropy alloys (HEAs) [1–6]. HEAs are generally located at the center of the phase diagram, it is anticipated to form many intermetallic compounds and complex phases. However, some HEAs form simple solid solution rather than many intermediate phases [1–5].

It is reported that the $\text{Ti}_{0.5}\text{CrFeCoNiCu}$ alloy exhibits high strength of up to 1650 MPa, together with extensive work hardening and large ductility of 22% [6]. However, Cu preferentially segregates to the interdendrite regions, and its effect on mechanical properties remains unknown. In this paper, more Al is added in the $\text{Ti}_{0.5}\text{CrFeCoNiAl}_x\text{Cu}_{1-x}$ alloy, and the phase transition and mechanical strength were investigated.

2. Experimentals

Ingots with nominal compositions of $\text{Ti}_{0.5}\text{CrFeCoNiAl}_x\text{Cu}_{1-x}$ (x values in molar ratio, $x=0, 0.25, 0.5$, and 0.75 , denoted by STC1, STC2, STC3, and STC4, respectively) were prepared by arc melting pure elements under argon atmosphere on a water-cooled Cu hearth. The alloys were remelted several times in order to improve homogeneity. Cylindrical rods with diameter of 5 mm were prepared by suction casting. Microstructure investigations of the cylindrical alloys were carried out by X-ray diffraction (XRD) using a MXP21 VAHF diffractometer with $\text{Cu K}\alpha$ radiation. The morphology and composition of as-cast specimens were examined with a LEO-1540 scanning electron microscope (SEM) with energy dispersive spectrometry (EDS). Cylindrical samples of $\phi 5 \text{ mm} \times 10 \text{ mm}$ were prepared for compressive tests and investigated with MTS 809 materials testing machine at room temperature with a strain rate of $1 \times 10^{-4} \text{ s}^{-1}$. A Cambridge S250MK2 scanning electron microscope was used to study the fracture surface of the compressive test samples.

3. Results and discussion

The XRD patterns of the as-cast alloys are presented in Fig. 1. The alloys of STC1 and STC2 show only the presence of FCC reflections. As the Al ratio increases to 0.5, a small amount of BCC reflections besides main FCC ones in STC3 alloy can be observed. When Al further increases to 0.75, two BCC reflections in STC4 alloy can be observed, one is with larger lattice parameter, the another is with smaller one. Clearly, with the increase of Al content, the phase gradually transforms from FCC to BCC. Interestingly, all the alloys form solid solution rather than intermetallics. The Gibbs free energy (G) is defined as

$$G = H - TS, \quad (1)$$

here H is the enthalpy, S is the entropy, T is absolute temperature. According to thermodynamics, the phases with lower G will form preferentially. If the enthalpy is constant, the phases with higher entropy will have the lower Gibbs free energy. The alloys of $\text{Ti}_{0.5}\text{CrFeCoNiAl}_x\text{Cu}_{1-x}$ contain seven components, and the entropy of mixing of all the alloys are higher than $1.7R$ (by using the regular solution approximation, and R is the gas constant), which is very high for the metallic alloys (because the entropy of melting of the metallic alloys is about $1R$). In addition, the entropy of the solid solution phase is much higher than that of the intermetallics, the former one is more random and the later one is more ordered. As a result, the Gibbs free energy of the solid solution phase is lower than that of the intermetallics, and so the solid solution phase preferentially forms, this entropy effect is largely enhanced for the high entropy alloys.

The superiority of simple solid solutions over intermetallics in the alloys is mainly attributed to the effect of high entropy of mixing [1,2,7]. By carefully studying the crystal peaks as shown in Fig. 1(b),

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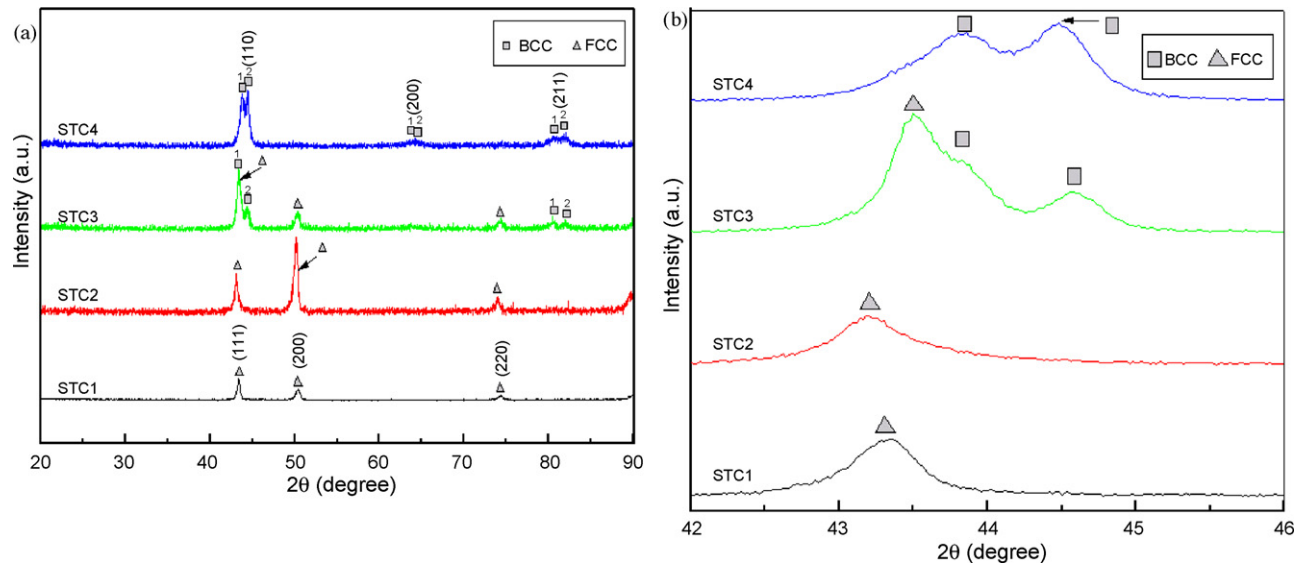


Fig. 1. XRD patterns of the $\text{Ti}_{0.5}\text{CrFeCoNiAl}_x\text{Cu}_{1-x}$ ($x=0, 0.25, 0.5$, and 0.75) rod samples with diameter of 5 mm and (b) is the detailed scans for the peak of (1 1 1) of FCC and (1 1 0) of BCC.

the peaks shift towards lower 2θ side with more Al additions for the STC1 and STC2 alloys. The left shifting of the reflections is attributed to the Al with larger atomic radius than any other atoms (except Ti), which will lead to large lattice distortion in the alloys. The lattice parameters of STC1 and STC2 are estimated to be 0.3612 nm, 0.3624 nm calculated from the (1 1 1) peak, respectively. When the BCC forms in the STC3 alloy as shown in Fig. 1(b), the (1 1 1) peak for the FCC shifts back the larger 2θ side. This indicates that the formation of the structure of lower atomic packing efficiency will decrease the lattice distortion energy.

Fig. 2 shows the SEM backscattered electron images of the alloys. All the alloys exhibit dendritic microstructure with different size and morphology. According to EDS results as listed in Table 1, the white phases segregating in the interdendritic regions in Fig. 2(a),

Table 1
EDS results of the white zones in Fig. 2(a), (b) and (c), respectively.

Region	Ti	Cr	Fe	Co	Ni	Cu	Al
(a)	1.19	1.94	2.20	2.53	3.93	88.21	0
(b)	6.10	4.86	7.53	6.39	10.23	59.95	4.94
(c)	3.38	4.49	6.01	6.41	12.31	59.04	8.36

(b), and (c), are identified as Cu-rich phase. The segregation is due to that Cu is more electronegative than other components and has an immiscible gap with Fe, Co, etc., thus is rejected into the interdendritic region when alloy solidified. In STC1 alloy, both dendritic and interdendritic phases gathered to clusters of about 2–3 μm . The EDS analysis reveals that Ni acts as the solvent and other atoms replace

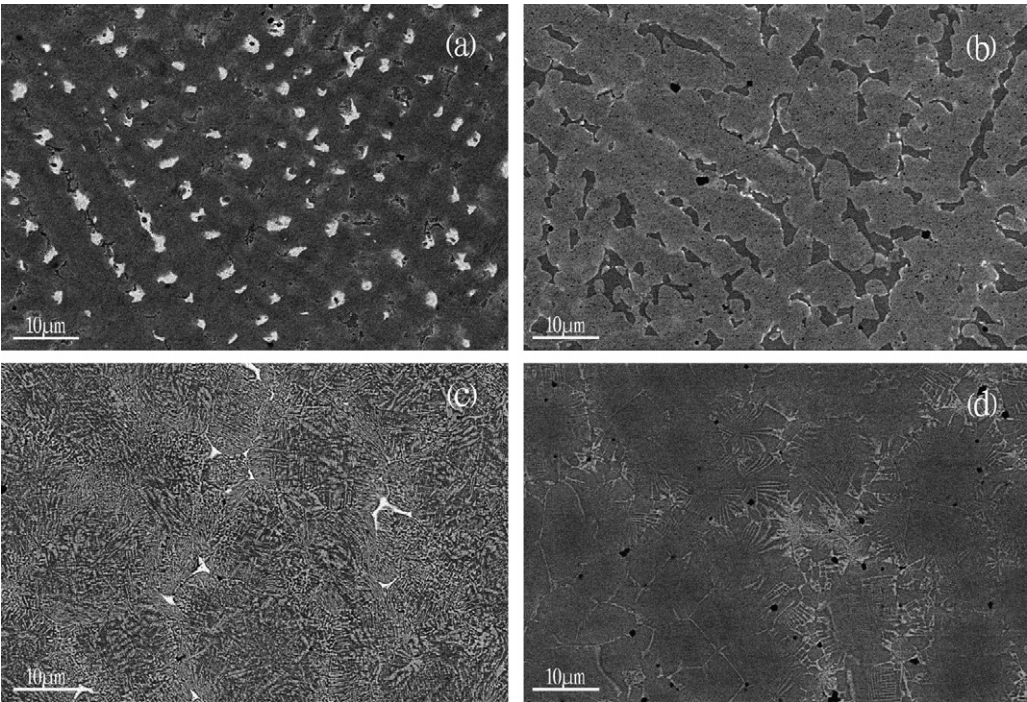


Fig. 2. SEM back-scatter electron images of $\text{Ti}_{0.5}\text{CrFeCoNiAl}_x\text{Cu}_{1-x}$ alloys after polishing, $x=0, 0.25, 0.5$, and 0.75 corresponding to (a), (b), (c) and (d), respectively.

Table 2

Room temperature compression test results for the alloy series.

Alloy	E (GPa)	ε_p (%)	$\sigma_{0.2}$ (MPa)	$\sigma_{\max}$ (MPa)
STC1	92.73	28.7	700	1650
STC2	102.85	38.5	750	1970
STC3	160.54	17.4	1580	2389
STC4	164.14	12	1900	2697
STC5	72.68	24.2	2040	3135

E , Young's modulus; ε_p , plastic strain; $\sigma_{0.2}$, yield strength; $\sigma_{\max}$, compressive strength; STC5, $\text{Ti}_{0.5}\text{CrFeCoNiAl}$ [4].

the Ni atoms in FCC lattice in STC1 alloy; While for STC2 and STC4 alloys, Ti is solvent in the black phases. As shown in Fig. 2(c), the overlapped lamellar structure indicates the mixture of FCC and BCC eutectic phases, which is consistent with the XRD patterns.

Fig. 3(a) shows the true stress–strain curves of room temperature compressive tests. STC1 and STC2 alloy exhibit similar high compressive strength and good plastic deformation. The further substitution of Cu by Al makes STC3 alloy has dramatically enhanced strength, but lower ductility. With Al content increasing to $x=0.75$, the maximum compressive strength increases to 2697 MPa, but the plastic deformation further decreases to 12%. The Young's modulus (E), plastic strain (ε_p), yield strength ($\sigma_{0.2}$), and compressive strength ($\sigma_{\max}$) are listed in Table 2. Fig. 3(b) shows the various tendencies of the compression test results for the alloys. Clearly, for $\text{Ti}_{0.5}\text{CrFeCoNiAl}_x\text{Cu}_{1-x}$ alloys, with the increase of Al content, the yield strength almost linearly increases, which can be ascribed to following: With the increasing of Al content, the lattice distortion energy will also be increased [8]. The lattice distortion will also enhance the effect of solid solution strengthening in the alloys, thus the strength will be increased.

Fig. 4 presents SEM morphologies of the fracture surfaces. For STC1 alloy, clusters of dendrites along various directions can be observed in Fig. 4(a), which might be attributed to its favorable ductility. High-density dimple-like fracture can be observed in Fig. 4(b), indicating that STC2 alloy has good plastic performance, which is in consistence with the compression test results. Fig. 4(c) shows the

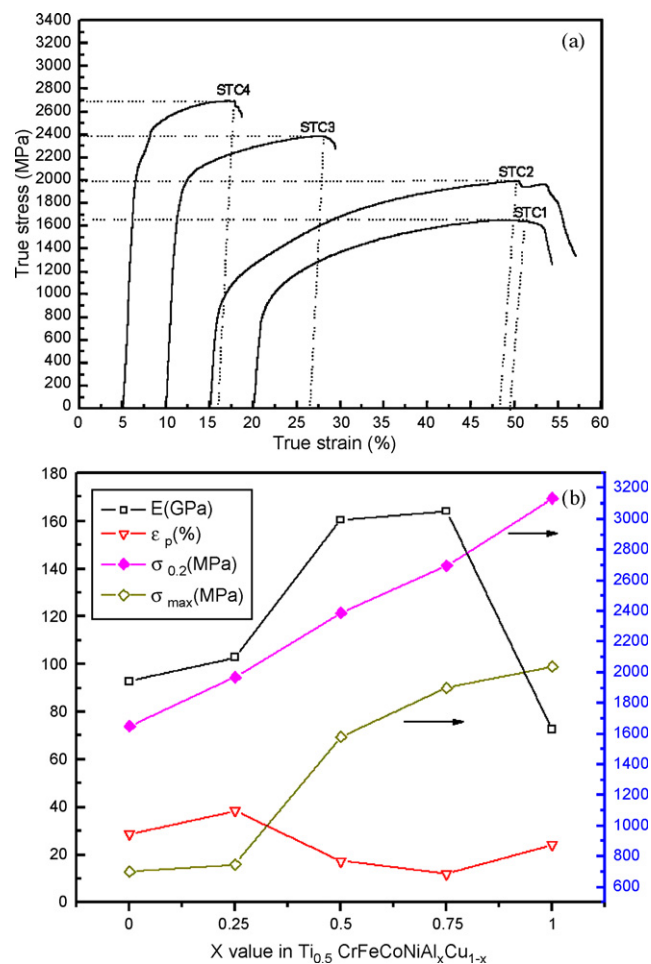


Fig. 3. (a) Compressive true stress–true strain curves of $\text{Ti}_{0.5}\text{CrFeCoNiAl}_x\text{Cu}_{1-x}$ alloys ($x=0, 0.25, 0.5$, and 0.75) alloys. (b) Compressive test results for the alloy series.

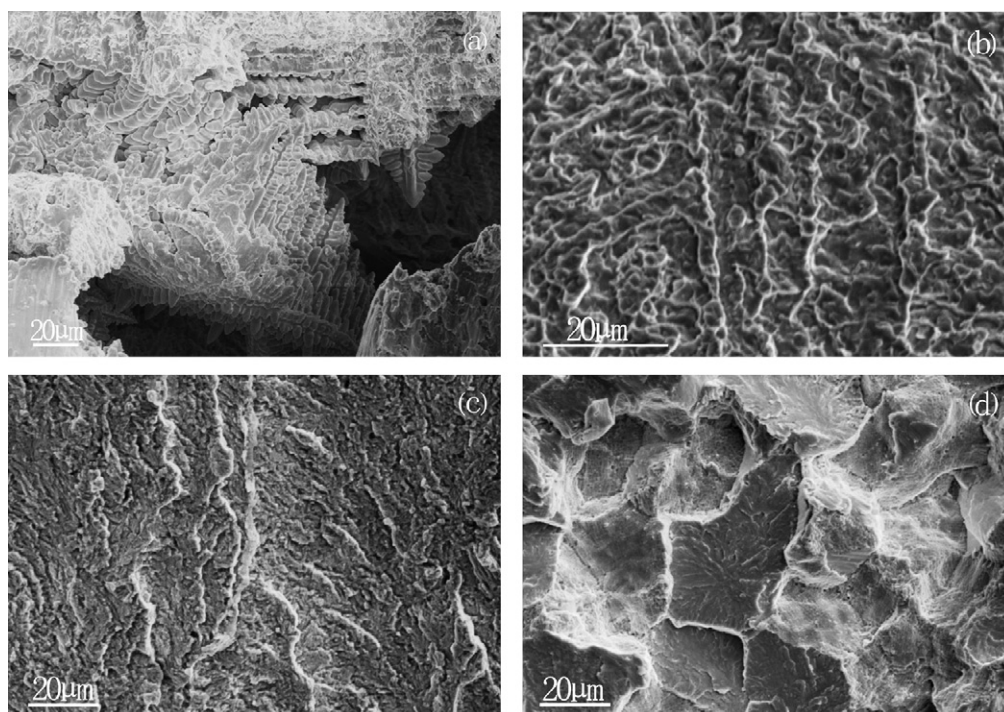


Fig. 4. SEM secondary electron images of $\text{Ti}_{0.5}\text{CrFeCoNiAl}_x\text{Cu}_{1-x}$ alloys, $x=0$ (a), $x=0.25$ (b), $x=0.5$ (c), and $x=0.75$ (d).

cleavage fracture of STC3 alloy. Corresponding to the FCC and BCC eutectic phases as shown in Fig. 2(c), the structure in the fracture surface is also fairly fine. On the surface of STC4 alloy in Fig. 4(d), cleavage steps and river-like patterns can be observed, which are of typical feature of cleavage fracture.

4. Conclusion

In summary, FCC or BCC solid solutions form in $\text{Ti}_{0.5}\text{CrFeCoNiAl}_x\text{Cu}_{1-x}$ alloys. With the increase of Al addition, the FCC phase gradually transit to BCC, and the compressive strength of the alloys also greatly increased from 1650 MPa to 2697 MPa. The phase transition and the increase of strength related to the atomic packing efficiency and the lattice distortion energy.

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